CSC4197

RESEARCH METHODOLOGY

LITERATURE MANAGEMENT, CITATION MANAGERS & REFERENCING

2 WHY CITATIONS EXIST

- Citations serve four functions:
  - **Credit** — acknowledging others' ideas (integrity)
  - **Evidence** — backing your claims
  - **Traceability** — readers can verify and dig deeper
  - **Positioning** — showing where your work sits in the field
- Failing any of these = plagiarism, weak arguments, or desk rejection.

3 CITATION STYLES IN CS

Style	In-text look	Used by
IEEE	[1], [2]–[4]	IEEE venues, most engineering
ACM	[Rahman et al. 2024] or [1]	ACM venues
APA	(Rahman & Das, 2024)	HCI, education, social-science-flavored CS
Harvard	(Rahman 2024)	Some journals, theses

Numbered styles save space (two-column papers); author-date helps readers recognize work
The venue dictates the style — your reference manager handles the conversion in one click.

4 ANATOMY OF A REFERENCE ENTRY

- A complete conference reference (IEEE):
- [3] M. Rahman, S. Das, and T. Khan, "Calibration-aware mortality prediction for ICU patients," in *Proc. IEEE Int. Conf. on Healthcare Informatics (ICHI)*, Dhaka, Bangladesh, 2025, pp. 112–121, doi: 10.1109/ICHI.2025.00123.
- Required fields: authors, title, venue, year, pages, **DOI**.
- Common student errors: missing DOI, citing the arXiv version of a published paper, "Available: google.com" URLs, inconsistent author-name formats.

5 REFERENCE MANAGERS: THE LANDSCAPE

Tool	Cost	Strengths
Zotero ⭐	Free, open source	Browser capture, Word & Google Docs plugins, group libraries, BibTeX export
Mendeley	Free (Elsevier)	PDF annotation, social features
EndNote	Paid	Institutional standard in some fields
JabRef	Free	Native BibTeX editor for LaTeX purists

We standardize on **Zotero** — everything you learn transfers to the others.

6 THE ZOTERO WORKFLOW

- **Capture:** browser connector icon on IEEE Xplore/ACM/Scholar page → metadata + PDF saved in one click
- **Organize:** collections per project; tags (`method:LSTM`, `to-read`); notes attached to items
- **Annotate:** built-in PDF reader — highlights become extractable notes
- **Cite in Word:** Zotero tab → Add Citation → bibliography auto-builds; switch IEEE↔APA instantly
- **Cite in LaTeX:** export collection → `refs.bib` (or Better-BibTeX auto-sync) → `\cite{rahman2025calibration}`
- **Collaborate:** Group Library shared with your project team

7 BIBTEX FOR LATEX USERS

- A `refs.bib` entry:

```
@inproceedings{rahman2025calibration,
  author    = {Rahman, Mahmud and Das, Sumi and Khan, Tareq},
  title     = {Calibration-Aware Mortality Prediction for ICU Patients},
  booktitle = {Proc. IEEE Int. Conf. Healthcare Informatics},
  year      = {2025},
  pages     = {112--121},
  doi       = {10.1109/ICHI.2025.00123}
}
```

- In the paper: `...as shown in \cite{rahman2025calibration}` → renders as [3].
- `\bibliographystyle{IEEEtran}` reformats everything automatically.
- ⚠️ Google Scholar's BibTeX export is often dirty — fix capitalization with {BERT} braces, verify venues. title = "Exploring BERT for language modeling“ → title = "Exploring {BERT} for language modeling"

8 CITATION ETHICS & HYGIENE

- **Cite what you actually read** — secondary citation ("as cited in") only when the original is unobtainable
- ❌ **Citation stuffing** — padding with irrelevant references
- ❌ **Coercive/self-citation abuse** — citing yourself/your supervisor without relevance
- ❌ **Fake references** — including **LLM-hallucinated citations** (Week 10!); every reference must be verified to exist, with correct DOI
- Cite datasets, software, and tools too (most have preferred citations)

9 RESEARCHER IDENTITY & METRICS

- **ORCID** — your permanent researcher ID (0000-000X-...); required by most journals; create yours today and let's do it now!
- **Google Scholar profile** — tracks your citations; set it up at graduation with your thesis
- **Metrics literacy:**
  - *h*-index = h papers with ≥ h citations each
  - Impact factor = journal-level average (says little about an individual paper)
- Use metrics as weak signals, never as the definition of quality — fields and career stages differ

10 CASE STUDY: THE RETRACTION THAT WASN'T CHECKED

- **Scenario:** A thesis cites a 2021 ML paper as core motivation. The paper was **retracted** in 2023 for data fabrication — the student never checked. An examiner notices in the viva.
- **Lessons:**
- Check retraction status for load-bearing citations (Retraction Watch database; publishers flag retractions on the page)
- Prefer citing the **published version** over preprints when both exist
- Zotero tip: the Retracted-item plugin flags retracted entries in your library automatically

11 HANDS-ON

- **Build your project's reference infrastructure:**
- (10 min) Capture your 8–10 Assignment-1 papers into a new Zotero collection via the browser connector — verify metadata of each (authors, year, DOI)
- (5 min) Create a **Group Library**; invite teammates + teacher (optional)
- (10 min) **Word path:** insert 3 citations + auto-bibliography in IEEE; switch to APA and back
- (10 min) **LaTeX path:** export collection as `refs.bib` → upload to your Week-8 Overleaf project → `\cite{}` three sources → compile
- **Due this week** (Intro + Related Work draft — now with proper citations!)

12 WRAP-UP & NEXT WEEK

- **Key takeaways**
- Citation style is venue-driven; managers make switching trivial
- Zotero workflow: capture → organize → annotate → cite → share
- Every reference must be **real, read, and correctly formatted** — hygiene is integrity
- **Next week:** AI in research writing, ethics & academic integrity — the most important class of the semester. Pre-reading: your university's plagiarism policy + one publisher AI policy (links on LMS)